

Supplementary Theory Note

Full Atomic Language and Finite Compatibility in Short Chain Families

Jihun Bae

Yeonho Bae

Jinglu Hu

This note accompanies “Exact Stabilization and Local Decision Structure in Shortest-Bridge Repair Dynamics.” It records the complementary N1/N2 atomic language and the finite short-chain compatibility theory. Notation and the Q2/Q3 mask-locality principle are as defined in the article.

1 Full Atomic Language

Definition 1.1 (N1 action). *Let Y be defective. Consider every disconnected occupied Q2/Q3 mask of Y and every unoccupied voxel v incident to at least one such mask. The singleton addition $\{v\}$ is an N1 action exactly when*

$$\text{pwc}(Y \cup \{v\}).$$

Equivalently, the N1 generator forms the union of the missing positions in all currently bad masks and retains precisely the one-voxel additions whose successors pass the exact Q2/Q3 predicate. If Y is valid, no N1 action is available.

Proposition 1.2 (Strict descent for the full atomic language). *Define*

$$\Phi(Y) = |\text{Comp}_6(Y)| + \mathbf{1}[\text{fail}(Y)].$$

Every admissible N1 or N2 transition $Y \rightarrow Y'$ satisfies $\Phi(Y') < \Phi(Y)$.

Proof. For an N1 action, the added voxel is drawn from a disconnected Q2/Q3 mask and repairs that mask in a globally valid successor. It is 6-adjacent to an occupied voxel of the current state, so the component count cannot increase, while the failure indicator drops from one to zero. For an N2 action, the retained path joins two distinct components, reducing the component count by at least one; the successor is valid by the N2 action definition in the article. $\square$

2 Finite Compatibility in Short Chain Families

Consider the interaction between a minimum-cardinality baseline repair and a second local repair in the family chain_L . The finite model has two complementary descriptions. Its minimum-cardinality repair family is represented by walks on four repair columns, while support separation reduces compatibility with the local repair to a fixed $3 \times 3 \times 3$ neighborhood. Together these descriptions give a three-layer compatibility bound, attained in the $L = 5$ family. The mathematical flow is

$$\begin{aligned} B \in \mathcal{M}_0(\text{chain}_L) &\xrightarrow{\text{walk representation}} \text{column walk}, \\ B &\xrightarrow{\text{localization}} B \cap \mathcal{L}(q) \xrightarrow{\text{three-layer span}} \text{pwc}(Z_{B,q}). \end{aligned}$$

For a defective voxel set F_0 , let

$$\begin{aligned} m_0(F_0) &= \min\{|R| : \text{pwc}(F_0 \triangle R)\}, \\ \mathcal{M}_0(F_0) &= \{R : \text{pwc}(F_0 \triangle R), \\ &\quad |R| = m_0(F_0)\}. \end{aligned}$$

Fix

$$\text{Seed} = \{(-1, -1, -1), (-1, -1, 0), (0, 0, -1), (0, 0, 0)\}$$

and, for $L \geq 3$, put

$$\text{chain}_L = \bigcup_{j=0}^{L-2} (\text{Seed} + (0, 0, j)).$$

At each occupied z -layer the four repair columns are $(-1, -1)$, $(0, 0)$, $(-1, 0)$, and $(0, -1)$; the two diagonally opposed pairs are $(-1, -1) \leftrightarrow (0, 0)$ and $(-1, 0) \leftrightarrow (0, -1)$.

A perturbation pair (a, b) is called *legitimate* in this finite setting when it is isolated from F_0 in Chebyshev distance at least two, each single perturbation preserves the complete minimal-repair family,

$$\mathcal{M}_0(F_0 \cup \{a\}) = \mathcal{M}_0(F_0 \cup \{b\}) = \mathcal{M}_0(F_0),$$

and the pair creates a defect witness absent from F_0 and from both singly perturbed states. Thus legitimacy combines isolation, individual invisibility, and joint visibility; equality of defect sets alone is not sufficient.

For such a pair let $F_{a,b} = F_0 \cup \{a, b\}$ and let q be an associated local repair voxel. Given $B \in \mathcal{M}_0(F_0)$, set

$$Z_{B,q} = F_{a,b} \triangle (B \cup \{q\}).$$

Compatibility means $\text{pwc}(Z_{B,q})$. Its *compatibility depth* is the smallest number of consecutive decision layers whose choices determine this verdict, with depth zero denoting a baseline-independent verdict.

Proposition 2.1 (Walk representation of minimum repairs). *For $L \in \{3, 4, 5\}$, the family $\mathcal{M}_0(\text{chain}_L)$ is in exact bijection with length- L walks on the four repair columns that never move between diagonally opposed columns at consecutive layers. Consequently,*

$$|\mathcal{M}_0(\text{chain}_L)| = 4 \cdot 3^{L-1}, \quad N_L(r) = 3^{L-r}$$

for every admissible prefix of length $r \in \{1, \dots, L-1\}$.

Proof. For $L = 3$, complete enumeration establishes minimality, the repair family, and the forbidden-diagonal transition rule. For $L = 4$, exhaustive rejection of all smaller candidate repairs establishes $m_0 = 4$; the completeness of the 108-member size-four family is then checked by the walk description and exact agreement with the finite reference classification. For $L = 5$, exact equality and P-well-composedness validity are checked for the 324-member walk family against the finite reference classification; minimality and completeness at this length use that complete finite classification rather than a second enumeration of all smaller and size-five candidates. Once the three finite bijections are established, the first layer has four choices and each subsequent layer has three legal continuations, yielding both formulas. $\square$

To state the locality result, let $\mathcal{N}(v)$ be the set of edge and vertex masks incident to a voxel v , let $C(h)$ be the voxels incident to a mask h , and put

$$\mathcal{L}(q) = \bigcup_{h \in \mathcal{N}(q)} C(h) = q + \{-1, 0, 1\}^3, \quad D_{\mathcal{M}} = \bigcup_{B \in \mathcal{M}_0(F_0)} B.$$

Lemma 2.2 (Locality under support separation). *Assume $q \notin D_{\mathcal{M}}$ and*

$$C(h) \cap D_{\mathcal{M}} = \emptyset \quad \text{for every } h \in (\mathcal{N}(a) \cup \mathcal{N}(b)) \setminus \mathcal{N}(q).$$

Then the verdict $\text{pwc}(Z_{B,q})$ depends on $B \in \mathcal{M}_0(F_0)$ only through $B \cap \mathcal{L}(q)$.

Proof. Every $F_0 \triangle B$ is locally clean. By the mask-locality lemma in the article, toggling a , b , and q can change only masks in $\mathcal{N}(a) \cup \mathcal{N}(b) \cup \mathcal{N}(q)$. On masks outside $\mathcal{N}(q)$ the displayed disjointness condition excludes every voxel whose occupancy can vary with B . The remaining masks lie in $\mathcal{N}(q)$, and their incident voxels have union $\mathcal{L}(q)$. Therefore two baselines agreeing on $B \cap \mathcal{L}(q)$ give the same local occupancies at every potentially affected mask and hence the same verdict. $\square$

Corollary 2.3 (Three-layer compatibility bound). *Under the hypotheses of Lemma 2.2, if every baseline repair selects one decision voxel per layer, then compatibility depth is at most three.*

Proof. Since $\mathcal{L}(q) = q + \{-1, 0, 1\}^3$, its intersection with any repair column is contained in the three levels $q_z - 1, q_z, q_z + 1$. Under the one-decision-per-layer property, $B \cap \mathcal{L}(q)$ is therefore determined by these three layer decisions. Lemma 2.2 then shows that the same three decisions determine compatibility. $\square$

Proposition 2.4 (Sharpness and two compatibility mechanisms). *For the distance-two candidate space of chain₅, the exact quotient has 76 symmetry classes representing 528 legitimate pairs. Its depth distribution contains four depth-three classes representing 12 pairs, so the bound of Corollary 2.3 is sharp in this domain.*

Proof. The finite classification enumerates all 528 legitimate distance-two pairs and partitions them into 76 symmetry classes. For each class representative, the compatibility verdict is evaluated on the complete 324-member walk family from Proposition 2.1. Comparing this truth table with the restrictions to consecutive decision windows of lengths zero, one, two, and three gives the least determining window, hence the compatibility depth. The resulting class counts are 64, 6, 2, 4 at depths 0, 1, 2, 3, while the corresponding orbit sizes sum to 468, 40, 8, 12 pairs. In particular, four classes representing 12 pairs require all three layers. Corollary 2.3 gives the matching upper bound, so depth three is attained and the bound is sharp. The complete orbit records are included in this Online Resource. $\square$

The depth-three classes realize two genuinely different predicates on the relevant decision window. In the *inclusion mechanism*, compatibility requires every decision in the window to occupy the prescribed column $(-1, -1)$, so the compatible subfamily is a prefix-restricted walk family. In the *exclusion/OR mechanism*, incompatibility is triggered whenever the column $(-1, 0)$ occurs at any position of the window; the predicate is therefore a union of layerwise exclusions rather than a conjunction of prescribed choices.

The 21 support-separation checks are exactly the nonzero-depth applicability instances: 3 for $L = 3$, 6 for $L = 4$, and 12 for $L = 5$. Depth-zero classes are excluded because their verdict is baseline-independent by definition. The complete orbit table, all representatives, the support-separation checks, and the mechanism witnesses are included in this Online Resource. The propositions above concern the stated finite chain families. The walk representation is established only for $L \in \{3, 4, 5\}$, and support separation is not asserted to hold automatically for arbitrary L . These finite compatibility results concern only the stated short-chain families and are not used in the proofs of the article's two main theorems.